

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AMA579
Subject Title	Learning Theory for AI
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	The objective of this course is to introduce students to the fundamental principles of learning theory that underpin modern artificial intelligence (AI) systems. The course covers core mathematical concepts, theoretical foundations, and recent advances, enabling students to understand, analyze, and evaluate learning algorithms from a rigorous perspective. Practical implications for the design and assessment of AI models will also be discussed.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) explain key mathematical foundations of statistical learning theory and their relevance to AI; b) describe and analyze popular learning paradigms, such as supervised, unsupervised, and reinforcement learning; c) evaluate the generalization ability of learning algorithms and understand the trade-offs involved; d) apply basic concepts of capacity, overfitting, underfitting, and regularization in AI models; e) critically assess theoretical guarantees and limitations of machine learning methods.
Subject Synopsis/ Indicative Syllabus	Key topics to be delivered: a) Learning Theory Foundations for AI: Motivation, history, and core concepts that underpin modern AI applications, from vision to language models. b) Supervised Learning and Generalization: Key ideas like empirical risk minimization, bias-variance trade-off, and generalization bounds, with relevance to training robust models. c) Model Complexity and Regularization: VC-dimension, Rademacher complexity, and regularization techniques, applied to control overfitting in real-world systems. d) Beyond Supervised Learning: Theoretical insights into unsupervised and reinforcement learning, linked to applications like clustering and decision-making. e) Advanced Topics and Limitations: Topics include PAC learning, uniform convergence, algorithmic stability, and theoretical gaps that

	highlight the current limitations of AI models. The module will also explore emerging research directions in theoretical deep learning.							
Teaching/Learning Methodology	<u>Lectures:</u> Lectures will introduce fundamental theoretical concepts, key results, and their practical implications for AI. Mathematical proofs will be presented where appropriate, with emphasis on intuition and applications.							
	<u>Tutorials:</u> Tutorials will reinforce theoretical understanding through problem sets, in-class exercises, and discussions. Selected readings and case analyses will be used to bridge theory and real-world AI applications.							
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks		% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
				a	b	c	d	e
	1. Problem sets / assignments		20%	✓	✓	✓	✓	✓
	2. Project		30%	✓	✓	✓	✓	
	3. Exam		50%	✓	✓	✓	✓	✓
	Total		100%					
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:							
	<u>Problem sets / assignments:</u> Students will solve theoretical and practical problems related to key concepts, derivations, and proofs. Assignments will foster deeper understanding of mathematical foundations and their application to AI, assessing students’ grasp of core concepts and ability to analyze learning algorithms.							
	<u>Project:</u> Students will complete a course project related to applying learning theory to a specific AI problem, demonstrating their ability to apply theory and algorithms, analyze results, and address real-world challenges.							
	<u>Exam</u> A written exam will assess overall understanding, analytical ability, and the capacity to synthesize and apply theoretical principles to novel problems in learning theory for AI.							
Student Study Effort Expected	Class contact:							
	▪ Lecture			26 Hrs.				
	▪ Tutorial			13 Hrs.				

	Other student study effort:	
	▪ Self-learning from reference materials	36 Hrs.
	▪ Reading and research for the case study	42 Hrs.
	Total student study effort	117 Hrs.
Reading List and References	[1] Shalev-Shwartz, S., & Ben-David, S., <i>Understanding Machine Learning: From Theory to Algorithms</i>, Cambridge University Press, 2014. [2] Mohri, M., Rostamizadeh, A., & Talwalkar, A., <i>Foundations of Machine Learning</i> (2nd Edition), MIT Press, 2018. [3] Vapnik, V., <i>Statistical Learning Theory</i>, Wiley-Interscience, 1998. [4] Bartlett, P., & Mendelson, S., <i>Rademacher and Gaussian Complexities: Risk Bounds and Structural Results</i>, in Handbook of Statistics, 2006. [5] Bousquet, O., & Elisseeff, A., <i>Stability and Generalization</i>, Journal of Machine Learning Research, 2002.	

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